

ActInf GuestStream 099.1 ~ Avel Guénin-Carlut: "From Contextuality to So

GuestStream | Avel Guénin-Carlut | 1:16:32

All right. Hello and welcome. It's 9th of March 2025. We're in Active Inference Guest Stream 99.1 with Avelle. So welcome, Avelle, if you would like to say hello and go into your presentation. We're looking forward to it. So go for it.

Hello. So I'm Avelle Guénard-Carlu. I work with Andy Clark, who is a cognitive philosopher at the University of Sussex in the context of the X-Kype Project, which is a project on archaeology, cognitive archaeology, which focuses on the relation between mind and natural culture. I'm also a founding member of Kyrus Scientific Para-Academic Organization. Think of it as a think tank. And what I will present today, let us consider it as an advertisement for the social science sub-project in the Active Inference Institute, because it's the aggregation of the work I did to motivate, well, necessity for, I will say, new mathematics for the study of social change that are inspired by Active Inference, but, you know, not really in line with what is done mathematically to formalize it. So we will go from contextuality to constraints, and I will try to articulate an Active Inference Ontology for participatory realism and social change. So, first let us try to motivate what is it that we are doing here. So, human minds and human culture display a very clear sign of open-ended evolution. What do I mean by open-ended evolution? It is evolution that is open-ended. I'm sorry, it's kind of circular, but essentially you can make new things things that did not exist before, that were not a thing before, that were not a part of some platonic space of things. We made it up. It was made up. This is what I mean by open-ended evolution. And this is true in generally whatever it is that you might do. So, for example, we have an impression of experience escape on the left of the screen. And, you know, it's buildings. We made them up. We built those buildings and they ensue a specific perception of what a building is that did not predate the concept of a building, the invention of building by humans. And this is very manifest, for example, in writing because, you know, writing uses sign. The interactivity of writing relies on signs being a thing. But signs are a thing only if we look at them as signs. So, we are looking at letters on this screen and want to stamp them as being embedded with meaning. But the concept of a C does not pre-exist humanity. There is no C dimension in the universal wave function at Big Bang. We built it up. It's a planet evolution. And the problem of explaining, presenting, addressing mathematically and physically the open-ended evolution of stuff is a pretty huge problem. Stuart Kaufman, best physics guy, in my opinion, dedicated his career to that. And, let's be frank, he did not solve it. He is not near to solving it. Because it's a hard problem. Because there is this idea that open-ended change brings out new stuff. And in mathematics, we have to represent a nice space that has specific possibilities in it. Else, we are not going far. And this seems to be incoherent with the logic of open-ended evolution. So, hard problem. I will jump to cognitive science because, again, my interest here is in human mind and social evolution. And I will try to explain what we mean by explaining in cognitive science.

So, for schematic purposes, I will say there are three kinds of explanations that we can go for. One is nomological. It's with regard to laws. So, for example, if I don't know a third thing and see that it falls this way, I can formulate a law that describes what happens. And because it happens a lot, happens every time I try, I claim it is a law. I claim that describing the regularity explains this activity. So, it's a phenomenological approach to explanation. I just described the phenomena and hope this is explanatory. This is how we do things in physics, not in psychology, not in biology, not in cognitive science, not in sociology. What is more natural in those disciplines, the special sciences, is a mechanical explanation. So, we look at structure and we show how the structure entails this phenomenon. For example, we have the Hodgkin-Luxley model of neural activity, which oscillates a bunch of electrochemistry, which, if verified, and it seems to be verified, explains the action potential activity of neurons. And I would say, pretty close to mechanical explanation in the space of possible explanation are functional explanations, which address not how does it work, but why is it? And that is centered on functional hypothesis. So, it's, for example, most of what biologists, evolutionary biologists do. It's adaptive hypothesis. So, why must this happen? Because it's selected for. And I would say, both, you can cluster those approaches between homological, where you tend to try to describe laws and mechanical and functional hypothesis, where you will have a more lean approach to modeling and trying to get into what is going on in a specific system for it to act like that. And in neuroscience, we are pretty good at mechanisms. This is, according to Brun and Piccinini, the hallmark of what they call cognitive neuroscience. We assume that the behavior, as we can be sure it, derives from cognitive mechanisms. Those cognitive mechanisms are implemented by the physical structure of the brain. And this physical structure is integrated in a context of operation that involves both physical, physiological and ecological factors. So, we can look at, for example, activation, patterns of activation in brain activity and map them into equation. Here, it's from Domathob. And it tries to map brain activity onto reinforcement learning, which is very very hype family. It was at least a very hype family of algorithm describing learning. So, we can explain activity from structure, but what we cannot do with the mechanistic approach is explain why the structure is there. Because you can describe, you can explain activity from structure, but if you try to explain structure from structure, you get, you know, regression to infinity. You do not explain much, or at least you do not have the possibility to explain without regressing to the infinite. The argument I will make here is that both cognitive structure, understood as, you know, patterns in cognitive activity, and its evolution flow from a single phenomena, which is the construction of meaning to cognitive agency. So, the structure produces a function, a pattern of activity, which itself will shade the structure at time t plus one. This is the only way I can think of to, you know, at least have a baseline concept of what it is to study upon the evolution. System does stuff, stuff changes system. Pretty simple. I don't see how we could, you know, go around that. And to be more precise, I will argue that we exist in a context that is characterized by social constraints and that define what we agree, let us say, we can and cannot do in a given structure context. And these

constraints are integrated and enacted by humans. And the way they are integrated and enacted, kind of, ontologically by construction leads to their reconstruction. And this is what underlies both the structure of cognition through the integration of thinking in those system constraints and the open-ended evolution thereof. So, we do active inference. This is the Active Inference Institute.

We may as well explain Active Inference. So, I guess most people who are attending to the talk now know what I'm talking about, but this is going on YouTube. I will reference it. I need to explain what Active Inference is.

Active Inference is built from what Carl Freestone and colleagues proposed as a principle for cognition, that is Bayes or Bayes and inference. Bayes and inference is basically reduces to the formula you see on screen.

The probability of an hypothesis given an evidence is equal to the probability of an evidence given an hypothesis times probability of the hypothesis over probability of evidence. So, for example, if you hear barking, it's pretty likely there is a dog. Unless it's very likely there is a dog near you. For example, you are in an office and the office is closed,

so there no dog can enter. Or unless you have an explanation for hearing barking. For example, you have a YouTube tab open with eight towers of barking noises.

I don't know why you would, but if you did, you would hear barking and nothing of a dog. So, this reduces to this formula.

P of E knowing H means the probability of hearing barking if there is a dog. Probability of H means the probability of having a dog with you.

Probability of E means the probability of hearing the barking. And you can compute the probability of having a dog with you knowing that you heard barking.

[illegible]

that could approach optimal Bayesian inference,
so the optimal inference of hypothesis given evidence
through minimization of something called
rational free energy.

You don't need to know exactly what this formula is.

What is important is that it's something that is always higher to surprise.

So if you minimize rational free energy,

you minimize surprise,
you approach perfect Bayesian model
of what is going on around you.
And if you assume that
your own activity layer upon layer upon layer
of your own activity self-organize like that
to infer the causes of sensory stimuli,
they will effectively predict upcoming signal
in a way that enables adaptive understanding
and navigation of the world.
And this is pretty much what the active inference paradigm is.
It's assuming that the brain does that.
We will now try to illustrate what exactly we mean by,
you know, active inference predictive processing.
The core idea is that your brain predicts sensation,
even prior knowledge out of the world.
So please look at the screen,
look at the sentence in the middle of it,
do not look away.
And pick something in your peripheral vision.
For example, I'm picking a water bottle at my right.
And while still looking at the screen,
you don't glance at the object you picked.
You try to make out as much detail as possible about it.
I'm going to predict two things for most of you.
You will be confident about knowing what it is.
You will see what it is,
but also you won't be able to see any single detail
that tells you what it is.
For example, I'm seeing water bottle.
I know it is water bottle,
but I just see a blue thing.
I do not have the detail to know it's water bottle.
It could be a vase for all I can see,
but I have the information that it is indeed a water bottle
because I put it there.
I'm remembering that if I want to drink,
I need to catch it.

So I know it's water bottle and I see water bottle.

You can now look away from the screen.

So what this illustrates is that your perception is at a pretty basic level embedded in parallel edge.

And this is something that we have known more or less intuitively since a pretty long time.

For example, this is the rabbit and dog illustration that Wittgenstein really liked.

And I will predict once again, that when you see this illustration, you see either a rabbit or a duck.

And you may, you know,

if I look at the,

what would be the back of the head of the rabbit

or the mouth of the, sorry,

the back of the head of the duck

or the mouth of the rabbit, I see the rabbit.

If I look at the ears slash beak, I see a duck,

but I always see one at a time.

I don't see, you know,

a complex 2D fields for which I have a duck value

and a rabbit value as a computational system would.

I see a rabbit or I see a duck.

And this is again,

because my perception is embedded in prior concepts

I have about the world.

And I know about rabbits, I know about duck,

and I will perceive either a rabbit or a duck,

knowing those are pretty much two different kinds of things.

I cannot perceive both.

So active inference entails that perception works,

as I said it would,

under a proactive processing,

but also that actions work the same.

So in a sense, under active inference,

agents will engage with the world

by maximizing the Bayesian coherence,

which is predicting, in a sense,

the kind of predicting we do
by minimizing rational free energy
of sensory motor states with their prior belief
through both perception and action.
And what this means is that I may, for example,
if I don't know about my,
what is the water bottle, I may glance at it.
But also if I want water, I will, you know, take it.
I will expect myself to solve the problem I have,
the fact I want water.
And this is this expectation that will lead
to the sending of motor signals
that enable me to grab the bottle.
This is a bit abstract because, you know,
cognitive neuroscience, it's mostly math.
But I want to make very clear why this matters.
In engineering, signal engineering, control engineering,
this kind of stuff, there is a very well known problem
that is called the inverse problem.
If you have a forward model of the world,
so how things move,
and even if this model is well known and well behaved,
directly inferring what is going on in the world
requires to work not with the forward model,
but with its inverse.
And most of the time, it's not feasible.
Let's take an example.
What do you see?
I will strongly expect that you see a big blurred thing
with red and orange.
This is a painting of William Blake,
the ancient of the days.
And just by applying blurring to it,
which is a very simple process, we all understand,
you can destroy the information about it
so that you cannot tell what is going on with the painting.
And this is in general a problem with any kind of inversion.
Even if you know the process that builds your sensation,

you do not necessarily know how to infer

what is going on from your sensation.

And the same goes for motor control.

Even if you know perfectly how your body moves,
you don't necessarily know what command to send
to create this or this or this consequence.

And the same goes for motor control.

Even if you know perfectly how your body moves,
you can calculate it.

And if you can calculate it, it will be very costly.

And it will be likely very fragile against both noise,
ambiguity, any kind of error in perception.

It will not, it will be very, very, very bad for adaptive control,
deciding on the run all the time.

Active inference, because agents do not compute explicit representation of the world,
but compute or process physically what they expect to sense
and what they expect to do given context.

They can perform adaptive control without being subject to inverse problem,
because there is no inversion.

There is just a problem model of what I see, what I do.

And this is why active inference is most likely correct, in my opinion.

And this is why it's very important in context that your actions flow from expectation
and not from explicit planning.

But this has non-trivial consequences for how we experience the world.

I don't know what I said, but it does have very obviously consequences for how we experience the world.

So because you anticipate in a joint manner both motor states and sensory states,
you will experience reality as a landscape affordances,
where affordances is a technical psychology term for opportunity for action.

So let's take again the water bottle.

The point would be that I do not only see it as, you know,
array of pixels that are blueish,

I mostly see it as opportunity for drinking.

And something that I force grabbing.

So I can grab it, I can drink for it.

And this is something that is very basic in my perception of the bottle.

Let's take another example.

So this is a path.

And if you look at the color, it's pretty much only green everywhere.

You cannot see a thing in the middle, but, you know,
from a purely colorimetric statistical perspective, it's not obvious.
But you see a path.
This is likely the first thing you saw about this picture.
And if I remove it, and I wait a while, ask you to remember what it is,
I predict that you will remember there is a path, and there is green, and there are trees.
And if I ask you to remember exactly which trees are where, you will not remember that.
Or you may have an idea of it, and now that I put it back,
you will see that your idea of it is likely false.
At least, I saw it work every time I could check in real life,
whether or not people remembered precisely the layout of the things.
The point is, you remember, you perceive from this picture,
mostly the fact you can walk in the center of it.
You perceive mostly an affordance.
And you also perceive basic personal concepts, green trees.
But you're keeping in mind only information that is critical for you to navigate it.
And with more information, you will not take it from an abstract model of everything that exists in the world.
You will just glance.
And this is from that that you have the impression of knowing a lot about your development.
Because when you don't know about something, you can just glance.
It's not because you have a full representation of everything you have seen up to this point.
This would, again, not be something computationally possible.
And what I take from this notion of experiencing the world as a landscape of audiences
is that it entails the dissolution of agency as we know it.
I have to say agency as we know it because if I just say agency, Andy will be upset
and I do not want Andy to be upset because he pays me.
So, I will say agency as we know it.
In the sense that you do not have a perception of the world that you take as an array of facts
for taking decision that you then apply.
Everything is embedded.
When you see something, you already see what you are going to do with it.
You already see the valence that it has for you, both in terms of action that you can do,
in terms of emotional consequences, etc. etc.
Your perception action cycles are not entailed by some kind of central executive agency.
Also, executive functions do play a role.
They flow with your self-model, what you expect yourself to do given context.
And this means that context pretty much decides for you, in a sense.
At least it can.

And this is from there that we build the act-in-france notion of social evolution slash social organization. The idea that was argued by Ramstel and by Vesier is that humans will integrate and shape expectations about agents such as themselves.

If I see people stopping at a stop sign, I will infer that you have to stop at a stop sign.

And that this is something that applies to me personally.

And this means that we develop a shared landscape of cultural affordances.

Affordances, again, mean possibilities for action.

And cultural affordances just mean that those are culturally constructed.

And those affordances, they are not just something that exists in some abstract space, that we think about when we want and that we can decide when we apply them and when we don't. They directly shape what we see about the world and what we expect to do.

So under active inference, what we actually do.

This is what I call embedded normativity.

And this entails again that we participatively construct a landscape that regulates or at least participate, regulate,

and at least contextualize every single action and perception that we can possibly do.

Everything.

And this sees a pretty strong result.

It seems that it can explain, you know, a lot about opening the evolution of culture.

But does it?

We will now take a look at the formal grounding of active inference.

The bit of mathematics that is understood to lend its most, if not all, of its legitimacy.

Which is called the free energy principle.

We are getting into the math stuff.

I'm sorry for lots of you that don't like it.

I promise it's interesting.

So if we assume a stochastic dynamical system.

Dynamical means it moves.

Stochastic means it has a random component.

And assuming that the system is partitioned between A, B, E.

An agent, a boundary, an environment.

And that the states of the agent and the environment are independent from each other.

Given those of B.

So all information you have about A.

Sorry.

All information that you could have in E about A, it's inside B.

Which is expressed in the equation in the middle of the screen.

I don't want to explain too much about it.

If you can read it, you can read it.

If not, we'll go back to something less mathematical or pretty soon.

Then there is what we call information geometry.

So geometry of the information that is in the system.

So, the system is in the system.

And given states of A, the agent, it gives you a belief, a distribution over the states of E.

And these beliefs, they minimize rational free energy.

So the thing that we said, that if it was verified, it would entail predictive processing at inference.

It is verified in visual case, given the statistical independence of two systems.

So pretty much everything does cognition in the inference sense.

Terms and conditions do apply to the statement.

So, first, when we talk about the Markov-Blanket, essentially what we mean is that there is individuation, there is an identity of the agent in the environment.

In the sense that it's a bit technical mathematically, but intuitively in the sense that they are separated by something that the agent can act on and the agent can sense.

For example, motor and sensory states.

And at least from an information-theoretic perspective, interaction is only mediated by states of the boundary.

And again, in the context of active inference and the Frenchie principle, it is taken as the criterion for the existence of A as an individual system.

Something that also is very important to this interactive inference thing is that the entire idea of predicting one's own action depends on a condition that we call a strange particle, where there is no direct observation possible of active states.

So, if I do not have a direct way to observe what I do, causally speaking, my internal states are not directly affected by my active states,

then you have this process of predicting one's own action, and then you have what Howey, a philosopher of acting inference, calls self-evidencing.

So, self-evidencing means that, in substance, an agent gathers proof of its own existence to action.

So, I hope you remember this notion of Bayesian coherence between a model and sensory motor states.

This is it.

If you maximize the Bayesian coherence between your prior beliefs, what you think the world is, and what you think you are in the world,

and what you actually experience, you're gathering proof for a model that corresponds pretty much directly to your own existence.

So, this is quite critical. It's the formation stone over which this entire active inference and frangery principle of philosophy lies. And so, I'm accentuating it a bit. What's important, mathematically, is the relation between this notion of model of the world and self-model,

and the active reaction and perception that an agent enacts.

So, we'll have to look a bit more at the mathematics.

I need to explain more mathematical things.

A Gauth's theory, sorry, is a mathematical theory where the dynamics of a system are caused by its geometry,

more specifically by symmetries in this system.

So, first we have to define the term of symmetries.

Symmetries are a family of transformation that do not affect the system.

So, for example, what you can see on the screen is the ceiling of the Isfar Mosque.

And what you can see is that you can just switch it a bit. I did not calculate how much precisely, but you can, you know, rotate it a bit.

And this won't change the picture. This won't change any observable property of the system.

This is what we call symmetry. And in this case, it is a discrete rotational symmetry.

You can have a lot of symmetry. The principle is you can apply a transformation to the system that notifies nothing about it.

And most physical forces that are known are Gauth's theories. Electricity is a Gauth's theory.

Weak force is a Gauth's theory. Strong force is a Gauth's theory, all of which are compatible in their description.

General relativity is another Gauth's theory, which is naturally compatible with others for a reason I do not, will not get into.

But in essence, you have a thing called the Einstein Hilbert tensor that describes the geometry of space-time, even the density of matter in it.

And this tensor is by construction invariant under coordinate change. You can move it around. It doesn't change the system.

This is what we mean by relativity. And it is this symmetry. This symmetry is sufficient to entail the entirety of relativity theory and the entire idea of

the geometry of space-time that causes change in the acceleration of particles. So this is neat. This is a very powerful mathematical tool that changes a complex equation into a

Oh, this is invariant. That's it. And there is an argument. There is a pretty strong argument that active inference is a Gauth's theory. And it is a Gauth's theory that

is entailed by the symmetries in the probability of presence of the system in a specific sub-part of the system.

I need to spend a bit of time over this. Actually, the math is only going worse from there. You are aware. So what you see on your screen, the picture, is the decomposition between the conservative and the disruptive components of the atom

flow of the atom's flow of the system. On the left. What you can see that is wobbling around is a conservative flow. So it's something that I don't need to explain the specific mathematics.

Essentially, this is what happens when you throw a rock and there is no exchange of energy. There is no evasible evolution of the system. And the disruptive component of the flow, it is on the contrary. What happens, the attraction toward a specific sub-site of the system is

the opposite of the system that will take energy to do its stuff. And when you can decompose an edidymics this way. And when you look at this system, what it does is that the dissipative component of the flow will attract the system toward the synchronization manifold, which is in a sense, the most likely trajectory of a system.

So for example, you will wake up, drink coffee, go to the restrooms, drive, walk to work, you should go to

work, I guess, work, go to lunch, etc.

This is how you do things. And if you skip some steps, like you forget to eat, especially in this neuro, you will be in a pretty bad state and you will won't be able to do your usual stuff anymore.

And there is a force that acts on the dynamics of your brain and that enforces states that are at least roughly coherent with this routine, most likely trajectory.

This is the dissipative component of the flow. And this is the gauge force entailed by the symmetries in the probability of presence of the system, the agent-environment system in its state space.

Okay, this was abstract. I'm sorry. And essentially, there is an argument made by myself and Sengupta and Tristan that because the dynamics of the system are pretty much set, at least within the ecosystem theory, this force reduces to the expression of attention given prior beliefs.

So, if I'm thirsty, that was not a setup. If I'm thirsty, I will pay attention to water. And as we said prior, attention perception is always laden with value. So let's laden with what you can do with something.

So I pay attention to the activity of opening water bottle and starting to drink.

I swear to you, this was not me setting up this. I was thirsty and just took the water bottle. I did not think about it. This was not me making a point.

So, I would say this is very nice. All of this is very math. I like math. But maybe we are running into a problem of circularity. Because under active inference, we accept that dynamics of the agent flow from the beliefs of the agent, their priors.

And more specifically, the symmetries in those priors, and more specifically, the symmetries in those priors beliefs. But also, in the context of the FEP, we take those beliefs to constitute the attract distribution of the agent-environment system, the actual probability of the system to go there and there.

So the beliefs are entailed by the dynamics, which are entailed by the beliefs. There is a circularity problem in that. Your definition of a system defines not only the system, but everything it can possibly do.

This means you cannot have this ribbon type phenomena, where your structure produces new activity, which shapes the structure in a novel way. Because again, everything is entailed by the symmetries of the system.

And you could say renormalization, you could say path integral, you could say a lot of words. But the FEPs define the context of the Nicholas system theory. This is what they do.

What they do. The Nicholas system theory is a theory where you have a space like dimension, x , you have a time like dimension, t , and you have a function ϕ that defines, in a sense, a flow in time.

So a flow in zero time is zero, and a flow in t_1 time, and then a flow in t_2 time is equal to a flow in t_1 plus t_2 time. This is just coherence condition that entail, that ensures this is actually a flow in time.

And this theory represents with maximal generality, the evolution of a system that is well defined, that has an actual embedding space, which evolves in time, along with a well defined flow, so that has physical dynamics, let us say.

And in essence, this framing is not general enough to describe open ended change, or so I will argue.

And here we go in the main course of this talk, the notion of contextuality in relation to active inference, core points I'm making.

So, in quantum physics, we have to talk about quantum physics, I'm sorry, I said it was only getting worse.

In quantum physics, there is a, you know, well defined physical space, but they are somewhat weird, and they are known to be somewhat weird.

So, we don't consider the state of system per se, we consider wave function, which are basically probability

philosophy that the FEP has and why it uses DST. So if the time dimension is commutative, so $T1 + T2$ is equal to $T2 + T1$.

then DST does not have contextuality. You can apply T1 plus T2 or T2 plus T1. These are the same thing. These give the same results. You could decide that the time dimension is non-commutative, but then you do not have a material understanding of time. The evolution of the system in the time dimension becomes a creative process where the agent could go any different way that he likes with different results.

So this is incoherent with the physical intuition that is behind DST as it's used by the FEP, which is to show the emergence of mind-line property beliefs from very simple physical property, clinical system property. If you just assume that your system is somehow creative by construction, you will get mind-line property, but you are not explaining it. So this entails a kind of conflict between Active Inference and the mathematical basis that it claims to have.

So Active Inference tells you that it grounds the description of cognitive agency, cognitive engagement, agency. This is the core claim of Active Inference.

But,

the fact that you have to do is take a bunch of neurons and just check them.

Berndt's synchronization, according to Markov-Blanket,

so a statistical in the appearance Blanket, as we defined.

And because of the FEP, we can say that this entails the implicit prediction of central inactive states, and this grounds cognitive agency.

This is the core claim of Active Inference.

So, it's not a classical UST, but classical UST does not allow to...

does not allow the description of a creative process where states of the world, like letters, take a new meaning and tell new dynamics, are made up along the way.

It doesn't give you contextuality, so it doesn't give you the constructive aspect of cognition, because this is the same thing.

Now, if we want to address this mathematically, we have to go in a kind of weirder, more esoteric direction. Specifically, what we get when we look at the thermodynamics of observation.

So, observation, when it is constrained by an agent's internal flow, has to bring about physical states, in the sense of well-defined observables that have a causal power on things.

The reason for this is not some weird Penrose or core stuff, it's just thermodynamics.

If you observe an object, if you are an agent S, and you observe an object O, you are getting information.

By Landau's principle, this entails heating.

So, this entails the creation of entropy.

This entropy has to go somewhere.

So, it has to destroy information.

This is what entropy is.

If you look at something, and you make it exist in the sense of some observable that has causal power, you have to destroy the possibility for observing something else.

So, meaningful observation, observation that you remember, because if you forget about it,

you do not have the same condition, entails the loss of information in the environment.

The lack of definition of what you are not looking at, because of either heating, forgetting, or quantum decoherence.

The reason why is developed in the Tegmark 2012 paper.

I won't explain the mathematics now, but I suggest you look at it.

And something that we can claim from it, if we are feeling adventurous, which I am, is following first, and the school of quantum Bayesianism, that you can pretty much reduce quantum cosmology.

So, the construction of physical space, as well as an end, relies critically on context.

You just need contextuality, understood as the construction of freely-dant observables from the choice of context by the agent, and build from that spaces.

So, in the 2013 paper that I am putting here, there is a demonstration of this in a specific subcase.

Again, I will not explain the mathematics.

I will just say that the mathematical work to show that it is possible is ongoing.

There is no definite demonstration that this is the math we need to use.

But, we have reason to believe this exists. There exists such math.

And, when we look at the role of our apologies, I forgot to mention this idea that physical spaces and physical observables are brought about in an untrivial sense by agency, by choice of context, by agent.

It is called perspatory realism.

And, it is core to the Bayesianist view of cosmology.

Bayesianist-like in France.

So, when we apply this notion of perspatory realism to humans, what I want to say, and again, I will have to be a bit short on the math,

is that assuming this thinking through other mind thing from earlier, assuming that humans think of themselves

as instance of a human type that has properties that has expected action,

then, and assuming contextuality, of course, then, successive cycles of observation or integration

and inaction of social constraints, understood as single expectation over what you are supposed to do,

entails, or at least enables the description of, a continuous process of upended reconstruction of cultural landscape.

And this, we know, is upended because of the contextuality which affords by construction

and then, when it changes, this is pretty much what it means.

And from there, we can have a dual understanding of the integration and inaction of constraints or landscapes that are built culturally and their evolution through the contextuality of the experience.

So, a single phenomenon of contextuality builds both community structure and its evolution.

This is nice.

So, let us try to sum up what was said here.

First, there is such a thing as participatory realism,

which claims that physical spaces and physical observables are built, brought about by the agency of

And this applies to the specific case of human social evolution by claiming that the relation between structure, so the dynamics of a given escape, and function, so the meaning it takes for individuals, is a core driver of main evolution. We have this ribbon-shaped thing where structure builds new stuff, which, sorry, builds new function, new meaning, that shapes structure in a constructive way. And this process cannot be described within DST because DST tells you what the structure is. It is what defines a chemical system, the flow it has. So, you cannot have evolution of flow within DST. It's not a thing.

And, from my own work and work of Frosse, I will claim that it translates quite directly in the contextual nature of cognition, the fact that we understand the world through concepts, and not through, you know, intrinsic perception of what's out there. This applies in particular to the case of human evolution, human social evolution, because we can describe assuming active inference, and assuming that contextual interpretation of it. The human social and cognitive activity can be described as the construction and direction of a shared system of social constraints, which shape public behavior and private thoughts through the expectation that they build over behavior and over attention. And this is an illustration of that process and the duality it entails between the and the action of structure and its reconstruction. And what's interesting with this concept is if you have constraints

And therefore, this entire contextuality thing not only explains the planet evolution in the context of social, social cultural evolution, but it also places this contextually at the center of, you know, the constructive aspect of life.

This entails that humans can be understood as a model, well, human social cultural life forms can be understood as a model for a broader understanding of life. And in that broader understanding of life, to build it.

[illegible]

[illegible]

so as to build certain meaning into this landscape.

And this enable ritual, the embodied synchronous activity

that is meant to have a specific function,

that is meant to entail specific norms and build specific emotions.

All of this is part of how humans set the context for social activity,

and all of this is stuff for which we have a lot of information.

So we can try at least to build mathematics to understand that.

And I wanted to conclude on that, saying by Wittgenstein,

Sarah saying that human agreement decides what is true and what is false.

It is what humans being say that is true and false,

and they agree in the language they use.

That is not an agreement in opinion, but in form of life.

Which is my opinion, the extent that we can understand interpret Wittgenstein,

and that is just a way to say that what is meaningful,

what is constraining in human activity is not the specific nature.

the literal meaning of a statement.

It's the choice of framings, and it's the context that is set by the language used by people.

Which is my opinion, a kind of visionary way to understand,

to preface all of this stuff.

So, I was of course not able to say a lot about the mathematics.

or really the mechanics that I suppose about all this stuff.

I had to stick to the basics.

And I think you would agree that basics were a bit abstract.

So I would strongly suggest that you look at the stuff I wrote about this.

I have two papers.

One on the right is Contextuality, Cognitive Engagement Act in France,

which is a quite bare-bones argument that contextuality is indeed intrinsic to cognition.

and critical to explain upon any change.

And the other on the left, from the existential stance to strong constraints,
is a thicker, more committed argument that indeed there are such things as social control from life,
that are entailed by social constraints, built by existential stance,
as understood under active inference.

So the entire program.

So thank you for listening.

I hope that it did not get too abstract, like moderate, acceptable onto exception.

And I guess I will take on the questions.

Thank you, Avil.

Awesome.

All right.

I'll start with writing with one question I wrote down,

and then look forward to anyone's questions in the live chat.

How do modeling and active inference techniques like structure learning,
dynamic Markov blanket detection, language-like notations for generative models,
deal or not with this question of the open-endedness of life and evolution?

Depends which technique and how it's applied.

I'd have to look at the fine prints.

But just to unpack why I'm asking, and then maybe it would be a little clearer,
is like you brought in this question of the open-endedness.

So what is the theoretical challenge posed?

Have prior methods been inadequate?

And what would be the case if there were some type of model flexible applications
that could track open-endedness or is that impossible?

I don't want to commit on possibility because I guess...

So all of this relies on the idea that the choice of state space is not natural.

It's built by a system with its activity.

So I don't want to commit to abstract possible or possible depends on what is interesting to you in the
process.

So for example, if I take structure learning, if you have some mechanism to postulate new concepts
through which a system will engage with the world, this is the kind of thing that drives the change.

My point is not that it's impossible to do.

My point is that it's not underwritten by the Franger Principle.

Because the Franger Principle is framed in the Nicholas Stem's theory,
where you have a fixed amount of stuff and a specific flow in the stuff space.

So if you want to do concept learning, you will have to postulate new things,
and it won't be as physically principle as it is in the FEP.

So what you would need to ground physically this kind of stuff is something else.

Or maybe a more specific, updated version of the FEP, I don't know.

But not something that is framed, strictly speaking, within ecosystem theory.

Awesome. Makes a lot of sense.

Okay, I'll ask a question from the chat. Anyone else can write.

Okay, Susan wrote.

Do social activities depend on individual allostatic capabilities?

Allostatic. Oh yeah.

Like, how do you say?

So the core argument, I would say, that is in the social constraints stuff, is that, let's say there is a coherence of what the constraints are.

between individuals.

So for example, the concept of money applies to you, it applies to me, it applies to anyone who lives pretty much in a urban context.

That does not mean that it applies the same to anyone.

Money won't mean the same for me or you, as it would mean for Elon Musk.

Because, for example, this is a, okay, this is, I'm getting technical on finance, but people who are really, really rich, like that are valued in billions, they do not actually have money on their bank accounts.

What they have is, are these actions, shares and stuff, which is valued billions, and they use it as leverage to get loans after loans after loans after loans.

So they get free money, basically.

This is not money that they spend.

So this does not mean the same thing when we say money, depending on whether we're considering the perspective of, you know, working class, or even normal person, a billionaire, a bank, a state.

All these agents can engage with money in different manner.

And you can make shit up, essentially.

You can decide that, no, I don't do a 10 euros bill.

You can invent new ways to think, you can, for example, you can build a common pool, a bank, basically with friends.

It's allowed, you can do it, you can build new meaning into things.

And this is something, I don't know if you see my cursor, but that is entailed by the integration of social constraints.

Because when I act in the context of constraints, okay, I'm constrained, these are constraints.

But I'm allowed to do with them, anything I can do with them, basically.

I'm allowed to project the meaning and to build the strategies that I like.

And this means that this system of constraints will change with time,

because people will find new way to use it to the advantage.

And yes, this is related to allostatic load.

This is related to pretty much anything, including a bodily signal, that is relevant for me to define my strategy.

The difficulty, again, is to represent mathematically how novelty occurs, because we can't just track every single state of everything.

And if we did, that still does not tell you where money is.

There is no money parameter in the big bank.

This is not a thing.

So money has been made up.

People have been made up.

Vision has been made up.

And the core difficulty that I'm trying to point out is to build mathematics that allow for this kind of construction of meaningful observable states.

And yes, again, agency is important, so allostasis.

How do you think that active inference picks up on that challenge of being able to model that?

So conceptually, there is the entire inactive semiotics thing.

So active inference allows you to think of cognitive engagements as something that's embedded in concepts.

So in agency and the context that agents project into the world.

And this is very good.

This is, I think, the way we should think about perception and action.

The problem is that the math does not follow.

The math does not math.

Again, we justify active inference.

There is a framework that is built within DST.

Contextuality, and therefore, change is not something that is afforded by DST.

There is a, so I assume someone in the chat is saying, yes, but what about what Chris Fields is doing?

Yeah, Chris Fields is doing strange categoristic stuff that has, I would say, the capability to show opportunity change because it has very few mathematical constraints and does not have the kind of constraints you find within set theoretical approaches.

I would say the issue is that you do not have this basic intuition of just better synchronization, building new stuff, because the mathematical framework is very general.

It's not just better synchronization anymore.

It's, I don't even know what it is.

It's info, how do you say?

Informorphism.

I don't know what informorphism is.

I have a definition in mind, but I do not know what to do with it.

So I would say there is a lot of work that remains to be done to build meaningful physical intuition about this process, because basically we have the choice between math that are too constrained to describe the change and math that is not constrained enough to explain it in terms of specific constraints and specific patterns of activity.

Yes, certainly that comes up.

Like what is the most general schema for modeling a cognitive agent or cultural agent?

It's just a blank spec.

The less the schema says, the more general it is, and the less is there, and the more degrees of freedom and modeler historiocity come into play.

Because the less there is, the more general it is, and then that's like where the low road and the high road kind of intersect is the free energy principle, which your talk also did a great job of bringing into and out of quickly.

Is architecture independent?

It's some relationships and frameworks that meet inactive inference with the bottom up, embodied, extended, et cetera, et cetera, lineage and ecological psychology, and all of the actual model constructivism and instrumentalism.

And it's, it's, it's, it's interesting how, yeah, you can, what do you think about this?

I'm not trying to do the, where you're doing that, I'm sorry.

Okay.

Jesse wrote, the problem with math and physics and any scientific discourse is the models just are more approximate of the error correction criteria.

The errors are more approximate of the error correction criteria.

Okay.

Where are you continuing on in your research and work?

Like after seeing things this way or is it laid out in the papers or where else are you going with it?

So I should really have done a part in the talk about the instrument.

So essentially I had to stop focusing on the math because I had to look at the sociology and I have to have to, I have to have a proof of work for this entire area of concepts.

And I'm focusing on this application to concrete process of what I call ground building.

So the construction of the set of priors that is shared by a community.

Mostly the two case studies that I've been focusing on are the Unification Church, which is Korean, far-right, Christian, light cult.

That's among other things, kept the conservative Japanese party in power for 70 years.

And it did the journal Reagan was reading and according to CIA, I did not verify, could a country in South America.

So as you can see, very, very important points for a cult.

And they are mostly known because of their ability to generate huge mobile panics around their coercive practices for recruitment and fundraising.

And so I wrote a paper about an active inference approach to this recruiting and conversion strategies, which was successful.

Because something we have with active inference and that is hard to address from a pure ethnographic perspective is the integration between linguistics, what we say, and the base cognition, how we experience something.

This is something that cognitive science, cognition is something that cognitive science studies pretty well, I think.

And the other major case study I've been working on is the construction of the landscape in the Sierra de Valenza, which is an archaeological site studied by Xscape, where you have a lot of mineral mounds that are playing around and who may or may not have a structuring role in the navigation of the area.

And so I've been doing theory and modeling on what's going on there.

That is very interesting.

Dare I ask how the mounds play a role in the navigation of the area?

So it is claimed that basically if you go from mound to mound, you are pretty close from an optimal path to travel through landscape, which is a very important point.

So it's not trivial to know where you are supposed to go.

I would say the non-trivial question is, at least in the context of the literature on it, did people intend to use the mounds as landmarks to help with navigation?

And I would say our core point is that intent, at least in the common sense, is not that important because from the moment those mounds are built, they do shape the landscape.

Like it does not matter when you say this or that what you profoundly mean in your inner self, because there is no such thing as our inner self.

There is what you do, the context in which you do it, and how it shapes the context.

And I would say this is a practical approach of ground building that I've tried to construct and illustrate.

Interesting.

It makes me think about sort of the first and third person biosemiotic perspectives.

Like when it comes to ant colony pheromone modifications of the environment or nest building, intention wouldn't come so much into play, yet it may dominate aspects of our experience.

So, considering, well, is the biosemiotic events of the pheromone perception or the mountain navigation or the reading or writing of an environmental mark, in what ways does the causal consequence or experience from the first person perspective accord with the outside behavioral observation from the third person perspective,

which sort of strips the experience of intentionality, yet does so with methods like intentional inference and inference to latent states proposed by modelers.

So clearly part of the modelers constraint, however, the reference of those modeling constraints is the biological constraint.

So, I guess this is something that's.

something that's pretty critical in the approach I'm proposing.

What we think of intentionality pretty much does not exist.

Because the baseline model that we have in our accidental studies, you have a body and you have a mind and the mind controls the body.

This is not a thing.

Well, you have a perception and use this perception to decide and act.

And this is not a thing.

Your perception of the environment is integrated to a perception of yourself, which is a direct cause for your behavior.

So, intentions are embedded in the environment.

And this is true for ants and the biosemiotics of your, you know, what's it called?

Well, pheromones.

And this is true for cults.

So, for example, there is an entire debate around are the munis, so the cult, the Unification Church cult, are they con-artists?

I don't know. I don't care.

I don't know if they believe in what they say.

I think they do.

I don't think that changes the morals of the situation because they have built the system and the system is bad.

And we do not have, we can just characterize the context, what it is, what makes people do.

And again, thinking about whether people intend to build something bad in the deep of their inner selves is not a meaningful question because there is no such thing as an inner self.

There is what you think to yourself.

It's a thing.

But there is no self that is not embedded in some context of operation.

This is not, if your self is what you expect yourself to do, how you expect yourself to behave, your self model, then it's always embedded in something, at least self-talk.

There is no way to escape that.

So again, there is no, let's say the notion of intention that we can motivate based on cognitive science, fields of cognition, it's much more minimal than the stuff that we expect and that we usually take to be necessary for responsibility.

So there is a problem.

And as far as I'm concerned, this problem can be solved from like sociologically by the release of premises, by the release of social constraints of the common grounds of social interaction.

And I would be pretty confident in it because if we can't do that, we can't do anything.

Cool.

Cool.

Cool.

Do you want to add any more comments or what else are you thinking about or how would you want to see people pick this up and continue?

Read my papers.

That's the simple stuff.

I'm trying to do an advert for them.

I'm trying to show evidence at least, but again, it's mostly math.

No one else in math by talking.

That's not a thing.

I don't think so at least.

So I would highly recommend that you take a look at the paper, which go in much more detail about, for example, the mechanics of social cognition as I understand them.

And I strongly suggest that you look into the social sciences project on the Active Inference Institute that I am founded, I guess I can say that.

And send me anything that you would want me to look at and I will look at it.

And I want to build new stuff from that.

This is the core message.

And I think it's pretty possible, pretty easy, pretty constructive, pretty productive, good things.

Awesome.

I want to ask so many questions about the social science project, but that's a great way to end it.

And I would look forward to it being reactivated this year too.

Yeah.

Let's do this.

Cool.

Thank you, Avel.

Thank you.

Bye.

Bye.

Bye.

Bye.